

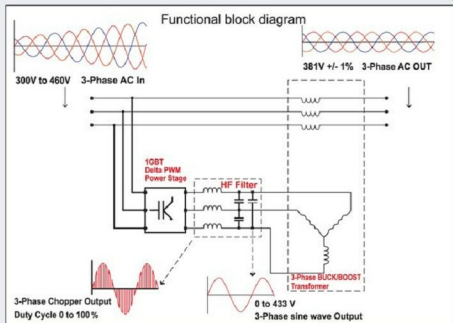
Technical know-how of Buck and boost Static Voltage Stabilizer Three-phase three-wire

MEI's three phase three wire buck and boost
Static Voltage Stabilizer using IGBT delta PWM.

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DSP based control PCB



Approximate Bill of Material (BOM) - Cabinet extra

Capacity (3ph)	Input Range	Output Voltage	Buck/boost Transformer VA	IGBT Current	Output Current	Bill of Material (BOM)
50KVA	330V to 430V (+/-13%)	380V	6.5KVA	11.4A	75A	Rs.50000
50KVA	310V to 450V (+/-18.5%)	380V	9.2KVA	17.2A	75A	Rs.58000
50KVA	300V to 460V (+/-21%)	380V	10.5KVA	20.2A	75A	Rs.60000
75KVA	330V to 430V (+/-13%)	380V	9.75KVA	17.1A	113.5A	Rs.62000
75KVA	310V to 450V (+/-18.5%)	380V	13.8KVA	25.8A	113.5A	Rs.66000
75KVA	300V to 460V (+/-21%)	380V	15.75KVA	30.3A	113.5A	Rs.68000
100KVA	330V to 430V (+/-13%)	380V	13KVA	22.8A	150A	Rs.68000
100KVA	310V to 450V (+/-18.5%)	380V	18.4KVA	34.4A	150A	Rs.72000
100KVA	300V to 460V (+/-21%)	380V	21KVA	40.4A	150A	Rs.75000
200KVA	360V to 440V (+/-10%)	400V	20KVA	32A	290A	Rs.80000
200KVA	330V to 430V (+/-13%)	380V	26KVA	45.6A	300A	Rs.90000
300KVA	360V to 440V (+/-10%)	400V	30KVA	48A	435A	Rs.105000

Technical know-how cost – Rs.18 lakhs.

Introductory offer "Rs.15 lakhs". Valid up to 31.01.2016

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- Three phase three wire operation. The PWM is done directly on Phase to Phase, not on neutral to phase.

Output is regulated for phase to phase voltage and hence for existing neutral to phase voltage also. So Neutral not required for the working of the unit. Single phase load can be connected from any output phase to existing neutral.

PWM type voltage regulation results smooth variation of the voltage and no need of voltage tappings.

- Neutral quality or availability is not affecting the output regulation or working of the unit.
- IGBT based PWM type voltage stabilizer which has tight regulation and fast correction speed. Output regulation of +/- 1% unachievable with SCR tap changing/servo stabilizers.
- Because of delta PWM only three half bridge IGBT modules required for the power stage
- Direct AC-AC conversion without rectifying to DC improves the efficiency, reliability and reduces the components
- Only the difference power is processed through the system resulting in small size buck-boost transformer and higher efficiency.
- 20KHz PWM control resulting silent operation and no distortion in output waveform.
- Automatic uninterruptable bypass in case of hazard
- Overload and short circuit protection provided
- Technology up to 200KVA three phase (up to 300KVA in +/-10% range)
- Four line LCD to display all parameters
- System has built-in data logger (optional). All parameters are periodically saved to flash memory with date and time stamp.